

Nouf Almesafri

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Education

MSc Applied Artificial Intelligence (2023-2024)

Cranfield University, Cranfield, UK

Related courses: Statistical Learning Methods, Search and Optimization, Deep Learning for Computer Vision, Data Analytics and Visualisation, Cyber-Physical Systems, Logic and Automotive Reasoning, Systems Engineering.

Grade: First-class degree

BSc Aerospace Engineering (2016-2022)

Khalifa University, Abu Dhabi, UAE

Related courses: Aerodynamics, Thermodynamics, Propulsion Systems, Dynamic Systems & Control, Space Dynamics and Control, Flight Dynamics and Stability.

Work Experience

Senior Associate Researcher (Feb 2024-Present)

Technology Innovation Institute, Abu Dhabi, UAE

Propulsion and Space Research Centre – Intelligent Engine Systems.

Experience in machine learning and feature engineering, including designing and training neural networks, anomaly detection and classification, and data analysis; complemented by physical model testing, pipeline creation, GUI development with PyQt and Qt Designer, and version control with GitHub.

Associate Researcher (Jan 2023-Feb 2024)

Technology Innovation Institute, Abu Dhabi, UAE

Propulsion and Space Research Centre – Aircraft Propulsion and Turbomachinery.

Experience includes designing a GUI, post-processing data acquisition, and engine testing.

R&D Trainee (June 2021-Aug 2021)

STRATA Manufacturing PJSC, Abu Dhabi, UAE

Experience includes robotic manufacturing applications, Vision-based software solutions, and testing.

Projects

Exploring the Impact of Noise on Vision Transformers and CNNs (Apr 2024-Sep 2024)

for Event-Based Object Detection

Simulated noise on an automotive dataset and fine-tuned ViT B16 and ResNet34 to achieve 86% and 88% accuracy respectively.

Urban Pollution Tracking with Drones

(Jan 2024-Apr 2024)

Developed a drone-based system using transfer learning and simulated gas emission scenarios to train the agent.

Vision-Based Approach for Robotics

(Jan 2021-Jan2022)

Developed an advanced solution using Python to perform manufacturing applications for aircraft.

URL: <https://www.ku.ac.ae/khalifa-university-and-strata-researchers-develop-vision-guided-cobots-for-high-precision-manufacturing-tasks>

Publications

Almesafri, N., Figueiredo, H., & Arana-Catania, M. (2025). From Ground to Air: Noise Robustness in Vision Transformers and CNNs for Event-Based Vehicle Classification with Potential UAV Applications. arXiv preprint arXiv:2506.22360.

Almesafri, N., Alhammadi, M., Zafar, S., & Santos, G. (2024, November). Power System Analysis of Solar and Hydrogen Fuel Cell Powered EVTOL UAV. In ASME International Mechanical Engineering Congress and Exposition (Vol. 88612, p. V003T05A006). American Society of Mechanical Engineers.

Almesafri, N., Alhammadi, M., Zafar, S., & Dos Santos, G. (2023, October). eVTOL UAV Conversion to Hydrogen Fuel-Cell Power Source for Enhanced Endurance. In ASME International Mechanical Engineering Congress and Exposition (Vol. 87615, p. V004T05A005). American Society of Mechanical Engineers.

Al Hammadi, M., AlMesafri, N., Zafar, S., & Santos, G. (2023, June). Design and Analytical Analyses of eVTOL UAV Performance Calculator for Power and Energy. In 2023 10th International Conference on Recent Advances in Air and Space Technologies (RAST) (pp. 1-6). IEEE.

Achievements

Certificate of Outstanding Paper

(Nov 2024)

10th IEEE International Smart Cities Conference on “Smart Cities: Revolution for Mankind”.

Dubai Airshow Presentation

(2021)

Presented vision-based approach for robotics project in STRATA booth.

Dean List

(2020)

First place in Khalifa Programming Contest

(2019)

Software Skills

Programming & ML: Python (PyTorch, TensorFlow, scikit-learn), Jupyter Notebook, Optuna

Data Handling & Visualization: NumPy, Pandas, Matplotlib, Plotly

GUI Development: wxPython, PyQt, Qt Designer

Collaboration & Version Control: Git, GitHub

Other Tools: LaTeX (Overleaf), Microsoft Office